IEOR 162 Project: Results and Analysis

Task 5: Sensitivity Analysis

Scenario	Profit / 1k Tickets	Demand Factor	Net Cost	Venues Opened	Bus Connections
A (baseline)	$10	1.0	$100,000	8	V1-V3, V2-V6, V7-V8, V9-V10
B (low profit)	$5	1.0	$1,504,000	8	V1-V3, V2-V6
C (high profit)	$15	1.0	-$1,315,000	8	V1-V3, V2-V6, V7-V8, V9-V10
D (overestimated demand)	$10	0.9	$363,200	8	V1-V3, V2-V9, V6-V8, V7-V10

5. Results and Analysis

5.1 Task 5 Results: Sensitivity Analysis

In Task 5, the Task 4 model is rerun under three alternative scenarios to test how sensitive the optimal solution is to changes in the assumed economic inputs. Two parameters are converted from fixed values to adjustable inputs. The first is the profit per thousand tickets sold, originally fixed at $10. The second is a demand multiplier that scales the per-session ticket demand for each sport. The baseline scenario A reproduces Task 4 exactly. Scenario B cuts profit per ticket in half to $5. Scenario C raises it to $15. Scenario D leaves profit at $10 but assumes the original demand estimates were 10% too high, so true demand is 0.9 times the listed value.

Each scenario is solved independently using the full Task 4 model. The optimal venue selection, sport assignment, weekly schedule, and bus network are recorded for each. All four scenarios open the same eight venues V1, V2, V3, V6, V7, V8, V9, and V10, and all four use the same sport-to-venue and week-by-week schedule. The only first-stage decision that responds to the scenarios is the bus network. Each scenario is solved using AMPL with the HiGHS solver, run from a dedicated `.run` file. The terminal output for each scenario is included immediately after its description.

Scenario A — Baseline (p = $10, demand_factor = 1.0)

Scenario A reproduces the Task 4 model exactly with no change to profit per ticket or demand. Solving this scenario establishes the reference plan: the same eight venues from Task 4 are opened, the same week-by-week schedule is used, and the four-pair bus network V1-V3, V2-V6, V7-V8, V9-V10 is selected. The optimal net cost is $100,000.

```
(.venv) rayyanscomputer@wifi-10-43-8-242 IEOR_162_Project % ampl run_task5_A.run
HiGHS 1.11.0: 100
HiGHS 1.11.0: optimal solution; objective 100
395840 simplex iterations
4618 branching nodes
absmipgap=4.26326e-14, relmipgap=0

"option abs_boundtol 3.4638958368304884e-14;"
or "option rel_boundtol 4.3298697960381105e-15;"
will change deduced dual values.


================ TASK 5 SCENARIO A (baseline: p=10, demand_factor=1.0) ================

Net Cost: 100.00 (thousand $)

Open Venues:
  V1
  V2
  V3
  V6
  V7
  V8
  V9
  V10

Bus Pairs Open:
  V1 - V3
  V2 - V6
  V7 - V8
  V9 - V10
```

Figure 1. AMPL terminal output for Scenario A (run_task5_A.run).

Scenario B — Low Profit (p = $5, demand_factor = 1.0)

Scenario B halves the profit per thousand tickets to $5 while keeping demand at the original estimate. With ticket revenue cut in half, two of the bus pairs no longer clear their $20,000 setup cost. The model keeps the same eight venues and the same schedule but drops the V7-V8 and V9-V10 bus pairs, leaving only V1-V3 and V2-V6 active. The optimal net cost rises sharply to $1,504,000.

```
(.venv) rayyanscomputer@wifi-10-43-8-242 IEOR_162_Project % ampl run_task5_B.run
HiGHS 1.11.0: optimal solution; objective 1503.999998
286480 simplex iterations
2396 branching nodes
absmipgap=0.0931456, relmipgap=6.19319e-05

================ TASK 5 SCENARIO B (low profit: p=5, demand_factor=1.0) ================

Net Cost: 1504.00 (thousand $)

Open Venues:
  V1
  V2
  V3
  V6
  V7
  V8
  V9
  V10

Bus Pairs Open:
  V1 - V3
  V2 - V6
```

Figure 2. AMPL terminal output for Scenario B (run_task5_B.run).

Scenario C — High Profit (p = $15, demand_factor = 1.0)

Scenario C raises the profit per thousand tickets to $15. Every bus pair clears its break-even threshold by an even wider margin, so the bus network is identical to the baseline. The same eight venues and schedule are opened. With $5 more revenue per thousand tickets sold across 283,000 effective tickets, the objective swings to a profit of $1,315,000 (a net cost of -$1,315,000).

```
(.venv) rayyanscomputer@wifi-10-43-8-242 IEOR_162_Project % ampl run_task5_C.run
================ TASK 5 SCENARIO C (high profit: p=15, demand_factor=1.0) ================

Net Cost: -1315.00 (thousand $)

Open Venues:
  V1
  V2
  V3
  V6
  V7
  V8
  V9
  V10

Bus Pairs Open:
  V1 - V3
  V2 - V6
  V7 - V8
  V9 - V10
```

Figure 3. AMPL terminal output for Scenario C (run_task5_C.run).

Scenario D — Overestimated Demand (p = $10, demand_factor = 0.9)

Scenario D keeps profit at $10 but assumes the original demand estimates were 10% too high, so the per-session ticket cap is scaled down to 0.9 times the listed value. Lower demand reduces extras per bus pair, but the model still finds four pairs that pay back at $10 profit. The optimal solution rewires the network to V1-V3, V2-V9, V6-V8, V7-V10, which yields the same uplift as the baseline under reduced demand. The net cost is $363,200.

```
(.venv) rayyanscomputer@wifi-10-43-8-242 IEOR_162_Project % ampl run_task5_D.run
HiGHS 1.11.0:
HiGHS 1.11.0: optimal solution; objective 363.199994
465338 simplex iterations
7619 branching nodes
absmipgap=0.0205458, relmipgap=5.6569e-05

"option abs_boundtol 2.3092638912203256e-14;"
or "option rel_boundtol 1.7270135938613546e-15;"
will change deduced dual values.


================ TASK 5 SCENARIO D (overestimated demand: p=10, demand_factor=0.9) ================

Net Cost: 363.20 (thousand $)

Open Venues:
  V1
  V2
  V3
  V6
  V7
  V8
  V9
  V10

Bus Pairs Open:
  V1 - V3
  V2 - V9
  V6 - V8
  V7 - V10
```

Figure 4. AMPL terminal output for Scenario D (run_task5_D.run).

Cross-Evaluation Matrix

To measure how robust each scenario's plan is, a 4x4 cross-evaluation matrix is constructed. In each cell, the first-stage decisions (venues, assignments, schedule, bus network) from the row scenario are fixed, and only the continuous ticket sales and bus passenger variables are re-optimized under the column scenario's profit and demand parameters. The diagonal of the matrix equals each scenario's own optimum and serves as a consistency check. Off-diagonal cells indicate the cost of committing to one scenario's plan when the world turns out to match a different scenario.

Plan from \ World is	A	B	C	D
A	$100,000	$1,515,000	-$1,315,000	$363,200
B	$118,000	$1,504,000	-$1,268,000	$376,200
C	$100,000	$1,515,000	-$1,315,000	$363,200
D	$100,000	$1,515,000	-$1,315,000	$363,200

The sub-optimality gap of each plan under each scenario is calculated by subtracting the column's diagonal value (the true optimum for that world) from each cell. A gap of zero indicates that the plan happens to be optimal in that scenario.

Plan from \ World is	A	B	C	D
A	0	$11,000	0	0
B	$18,000	0	$47,000	$13,000
C	0	$11,000	0	0
D	0	$11,000	0	0

Plan A is optimal under A, C, and D, and only $11,000 worse than optimal under B. Plan B is the worst-performing plan in every other world, with a gap of $47,000 under high profit. Plans C and D match A's robustness exactly.

5.2 Task 5 Analysis

The most important result of Task 5 is that facility selection is completely invariant across all four scenarios. Every scenario opens the same eight venues V1, V2, V3, V6, V7, V8, V9, and V10, and the total fixed venue cost remains $2,850,000 in every case. This is not a coincidence. The venue opening decision is forced by structural constraints rather than economic ones. The model must schedule 19 required sport-weeks across venues that each cap at two or three sport-weeks, and each sport can only be hosted in venues that match its eligibility. Once those constraints are imposed, no combination of the eight smaller venues can absorb all required sessions, and the eight chosen venues are the cheapest combination that can. Changing the profit per ticket or shaving demand by 10% does not affect this arithmetic, because profit and demand only enter the model through ticket revenue, not through scheduling capacity.

The week-by-week schedule is also invariant across scenarios. Sport-to-venue and sport-to-week assignments are fixed by the same eligibility and capacity constraints that govern venue selection. The optimizer has no incentive to move sports between venues or weeks unless doing so improves bus revenue, and rearranging sports to support a different bus network would always either reduce ticket revenue or violate eligibility.

The only first-stage decision that responds to the scenarios is the bus network. Each candidate bus pair has an effective break-even profit per ticket, defined as the bus setup cost of $20,000 divided by the extra tickets the pair generates over the three-week period. The four pairs in the baseline plan A have very different break-even points. The V1-V3 pair generates approximately 11,000 extra tickets, so it breaks even at about $1.82 per thousand. The V2-V6 pair generates approximately 8,100 extras and breaks even at about $2.47. The V7-V8 pair generates only about 2,700 extras and breaks even at about $7.41. The V9-V10 pair generates about 3,200 extras and breaks even at about $6.25.

The pattern of bus changes across scenarios follows directly from these break-even points. At the baseline of $10 per ticket, all four pairs clear their break-even threshold and are profitable. At the high profit of $15, every pair clears with even more margin and the plan does not change. At the low profit of $5, only V1-V3 and V2-V6 clear their break-even, while V7-V8 and V9-V10 fall below it. The model therefore drops those two pairs in scenario B, saving $40,000 of bus setup but forgoing roughly $30,000 of bus revenue, for a net gain of about $10,000 over keeping them. The 10% demand reduction in scenario D reduces the extra tickets each pair can generate, but the model finds an alternative four-pair configuration V1-V3, V2-V9, V6-V8, V7-V10 that yields the same total uplift as the baseline. This is a degenerate optimum in which two different bus networks produce the same objective value, and either configuration is acceptable.

The cross-evaluation matrix shows that plan A is the most robust choice for the committee. Plan A is optimal under both high profit and overestimated demand, and is only $11,000 below optimal under low profit. The same is true of plans C and D, which match A's robustness because they include the same two high-revenue bus pairs V1-V3 and V2-V6 plus two additional pairs that pay

back at $10 or higher. Plan B is the least robust, because skipping V7-V8 and V9-V10 saves $40,000 in setup costs in the low-profit world but forgoes substantially more revenue if profit recovers to baseline or rises to the high case. Applying plan B in the high-profit world is $47,000 worse than the optimal plan for that world, the largest sub-optimality gap in the matrix.

The key planning conclusion is that the Task 4 plan is robust to the kinds of economic uncertainty studied in Task 5. The committee can commit to opening the eight venues, to the sport-to-venue assignment, and to the weekly schedule without concern that changes in ticket profitability or modest demand overestimates would invalidate those decisions. The two large bus pairs V1-V3 and V2-V6 are also safe commitments, since they remain profitable even at $5 per ticket. The two smaller bus pairs V7-V8 and V9-V10 are conditional on profit per ticket remaining at roughly $7 or higher, and would only need to be reconsidered if margins are expected to fall meaningfully below that level. Adopting the Task 4 plan in full is therefore the recommended decision, since its worst-case loss across all three alternative scenarios is only $11,000.